12.1 SSDs can be used as a storage layer between memory and magnetic disks, with some parts of the database (e.g., some relations) stored on SSDs and the rest on magnetic disks. Alternatively, SSDs can be used as a buffer or cache for magnetic disks; frequently used blocks would reside on the SSD layer, while infrequently used blocks would reside on magnetic disk.

a. Which of the two alternatives would you choose if you need to support real-time queries that must be answered within a guaranteed short period of time? Explain why.

b. Which of the two alternatives would you choose if you had a very large *customer* relation, where only some disk blocks of the relation are accessed frequently, with other blocks rarely accessed.

a. **选择SSD作为存储层更好**

- 直接将数据存储在SSD上可以保证数据始终从高速设备读取，不存在数据存储的分层，节省数据搬运的时间
- 如果SSD作为缓存，某些请求可能仍然需要从低级存储中读取，会导致搬运数据的延迟

b. **选择SSD作为缓存更好**

- 表非常大时，无法将整张表都放在SSD中
- 在更高的存储层次中无法提前预知哪些数据会被频繁访问，需要通过分层的方式将频繁被用到的数据过滤到高层次的存储器
- 缓存机制可以自动将频繁访问的块保留在SSD上

13.5 It is important to be able to quickly find out if a block is present in the buffer, and if so where in the buffer it resides. Given that database buffer sizes are very large, what (in-memory) data structure would you use for this task?

Hash table:当我们在维护较大的databse buffer时，快速判断数据是否存在并找到具体位置更好的方式是通过计算的方式，而不是遍历的方式。

- 堆文件组织：是一种无序的存储方式，记录可以放在文件的任何地方，寻找某一数据需要遍历整个文件，速度缓慢
- 顺序文件组织：一种有序的存储方式，记录按照一定的顺序排列，但寻找某一数据需要将对应数据与搜索键进行比较搜索而不能直接定位
- 多表聚集文件组织：判断数据的是否存在并找到具体位置时没有必要同时关注多个关系
- B+树文件组织：哈希表的查询速度更快，时间为O(1)，而B+树的查询速度更慢，时间为

O(log n)

13.9 In the variable-length record representation, a null bitmap is used to indicate if an attribute has the null value.

a. For variable-length fields, if the value is null, what would be stored in the offset and length fields?

b. In some applications, tuples have a very large number of attributes, most of which are null. Can you modify the record representation such that the only overhead for a null attribute is the single bit in the null bitmap?

a. 当可变长度字段的值为NULL时，offset字段可以置为-1(作为一个标记值，当读取到-1默认为NULL)，长度设置为0，这也是符合实际情况的

b. 我们可以将`null bitmap`表放在记录的开头

- 定长属性，我们直接存储值
- 变长属性：存储o表中的ffset和length两个字段信息

所有属性的信息都是按照初始顺序存储在`null bitmap`

- 在进行访问时，首先读取`null bitmap`确定哪些属性是非NULL的，然后按照`null bitmap`中的存储顺序对 非NULL属性进行访问

13.11 List two advantages and two disadvantages of each of the following strategies for storing a relational database:

a. Store each relation in one file.

b. Store multiple relations (perhaps even the entire database) in one file.

a.

- advantage
  - 可以针对每个关系的特点选择最合适的文件组织方式
  - 单个关系的操作不会影响到其他关系
- disadvantage
  - 夸关系访问需要访问多个人件，连接操作效率较低
  - 管理难度较大，需要维护大量文件，系统元数据管理开销增加

b.

- advantage
  - 连接操作高效，相关数据可以在物理上聚集存储，减少跨文件访问的开销
  - 管理方便，文件数量少，事务处理(尤其是涉及多关系的)处理更直接
- disadvantage
  - 灵活性差：所有关系必须使用相同的存储结构，难以针对单个关系进行优化
  - 隔离性问题：单个文件损毁可能影响多个关系